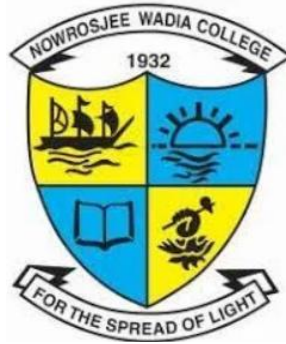


**Modern Education Society's
Nowrosjee Wadia College,Pune(Autonomous)**



**Department of Computer Science
T.Y.B.Sc(Information Technology)(NEP 2.0)**

**MINI PROJECT REPORT
ON
“BIKE RENTAL DEMAND PREDICTION”
A Supervised Learning Project Using Statistical Analysis and Linear
Regression**

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1. Introduction

Bike-sharing systems have become an important part of urban transportation, providing people with an affordable, convenient, and environmentally friendly way to travel. The demand for rental bicycles is not constant and can change depending on several factors such as time of day, weather conditions, season, holidays, and working days. Understanding these factors can help bike-sharing service providers manage the availability of bicycles and improve the efficiency of their services.

This project focuses on predicting hourly bike rental demand using the Seoul Bike Sharing Demand dataset. The dataset contains information about the number of rented bikes along with weather conditions and other time-related and seasonal factors. Statistical techniques and exploratory data analysis will first be used to understand the dataset and identify important patterns. A supervised machine learning regression model will then be developed to predict the number of bikes rented based on the available factors.

The project aims to demonstrate how statistical analysis and machine learning can be combined to solve a real-world prediction problem and provide meaningful insights into bike-sharing demand.

2. Problem Statement

Bike rental demand varies throughout the day and is influenced by several factors such as temperature, humidity, wind speed, rainfall, snowfall, season, holidays, and whether the day is a functioning working day. Because of these variations, it can be difficult for a bike-sharing service to determine how many bicycles may be required at a particular time.

The problem addressed in this project is to predict the number of bikes rented per hour using historical bike rental, weather, and time-related data. The Seoul Bike Sharing Demand dataset is used for this purpose.

Statistical analysis and exploratory data analysis will be performed to understand the distribution of bike rentals and the relationships between demand and different factors. A supervised machine learning regression model will then be developed using relevant input variables to predict the hourly number of rented bikes.

The project will also evaluate the performance of the model using appropriate regression evaluation measures and interpret the results to understand which factors are associated with changes in bike rental demand.

3. Objectives

The main objectives of this project are:

1. To study and understand the Seoul Bike Sharing Demand dataset and identify the important variables related to bike rental demand.
2. To perform statistical analysis on the bike rental data using measures such as mean, median, standard deviation, quartiles, and correlation.
3. To explore the relationship between bike rental demand and different factors such as temperature, humidity, rainfall, season, hour, holidays, and functioning days.
4. To visualize and identify patterns and trends in hourly bike rental demand using suitable graphs and charts.
5. To develop a supervised machine learning regression model for predicting the number of bikes rented per hour.
6. To evaluate the performance of the regression model using appropriate measures such as MAE, MSE, RMSE, and R^2 score.
7. To interpret the results and identify the factors that are associated with variations in bike rental demand.
8. To understand the practical use of statistical analysis and machine learning in predicting bike-sharing demand and supporting better resource planning.

4. Dataset Description

The dataset used for this project is the Seoul Bike Sharing Demand Dataset. It contains hourly information about the number of rental bikes used in Seoul, South Korea, along with various weather conditions, time-related factors, and seasonal information. The dataset is used to study the factors affecting bike rental demand and to predict the number of bikes rented per hour.

The dataset is obtained from the UCI Machine Learning Repository and contains 8,760 observations and 14 variables. Each observation represents the bike rental information recorded for one hour. The target variable for this project is Rented Bike Count, which represents the number of bikes rented during a particular hour.

Dataset Information:

Description	Details
Dataset Name	Seoul Bike Sharing Demand
File Name	"SeoulBikeData.csv"
Source	UCI Machine Learning Repository
Number of Observations	8,760
Number of Variables	14
Target Variable	Rented Bike Count
Problem Type	Regression
Missing Values	No missing values

Variables in the Dataset:

Variable	Description	Type
Date	Date on which the observation was recorded	Date
Rented Bike Count	Number of bikes rented during the hour	Numerical
Hour	Hour of the day	Numerical
Temperature	Temperature recorded in °C	Numerical
Humidity	Humidity percentage	Numerical
Wind speed	Wind speed recorded in m/s	Numerical
Visibility	Visibility measured in 10 m	Numerical
Dew point temperature	Dew point temperature in °C	Numerical
Solar Radiation	Solar radiation measured in MJ/m ²	Numerical
Rainfall	Rainfall measured in mm	Numerical
Snowfall	Snowfall measured in cm	Numerical
Seasons	Season during which the observation was recorded	Categorical
Holiday	Indicates whether the day was a holiday	Categorical
Functioning Day	Indicates whether the bike-sharing system was functioning	Categorical

Target and Predictor Variables

The target variable in this project is 'Rented Bike Count'. It is the variable that the supervised learning model will attempt to predict.

The remaining variables, including weather conditions, hour, season, holiday status, and functioning-day status, are treated as predictor variables. These variables will be analyzed to understand their relationship with bike rental demand and their usefulness in predicting the target variable.

The dataset is appropriate for this project because the target variable is numerical and the project aims to predict its value using other available variables. Therefore, the problem is treated as a supervised learning regression problem.

5. Data Preprocessing:

Data preprocessing was performed to understand the structure and quality of the Seoul Bike Sharing Demand dataset before carrying out statistical analysis and machine learning.

5.1 Loading the Dataset:

The SeoulBikeData.csv dataset was loaded into Python using the Pandas library. The dataset was examined to identify its number of observations, variables, column names, and data types.

5.2 Checking Dataset Dimensions:

The dataset contains 8,760 observations and 14 variables. Each observation represents bike rental information recorded for one hour.

5.3 Checking Missing Values:

The dataset was checked for missing or null values. No missing values were found, so no missing-value imputation or removal was required.

5.4 Identifying Variable Types:

The variables were examined and classified into numerical, categorical, and date-related variables.

Numerical variables include:

1. Rented Bike Count
2. Hour
3. Temperature
4. Humidity
5. Wind speed
6. Visibility
7. Dew point temperature

8. Solar Radiation
9. Rainfall
10. Snowfall
11. Categorical variables include:
12. Seasons
13. Holiday
14. Functioning Day

The Date variable represents the date of each observation.

5.5 Checking the Target Variable:

The target variable selected for the project is Rented Bike Count because the objective is to predict the number of bikes rented during an hour.

5.6 Checking for Outliers:

Numerical variables, particularly Rented Bike Count and weather-related variables, were examined using statistical measures and graphical techniques such as box plots. Extreme observations were not automatically removed because they may represent genuine periods of high rental demand or unusual weather conditions.

5.7 Preparing Variables for Analysis:

For the initial Linear Regression model, the numerical predictor variables were selected:

1. Hour
2. Temperature
3. Humidity
4. Wind speed
5. Visibility
6. Dew point temperature
7. Solar Radiation
8. Rainfall
9. Snowfall

The target variable Rented Bike Count was separated from these predictor variables. The prepared data was then used for Exploratory Data Analysis, statistical analysis, and Linear Regression model building.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution, patterns, and relationships present in the Seoul Bike Sharing Demand dataset. Statistical measures and graphical techniques were used to examine bike rental demand and its relationship with different factors.

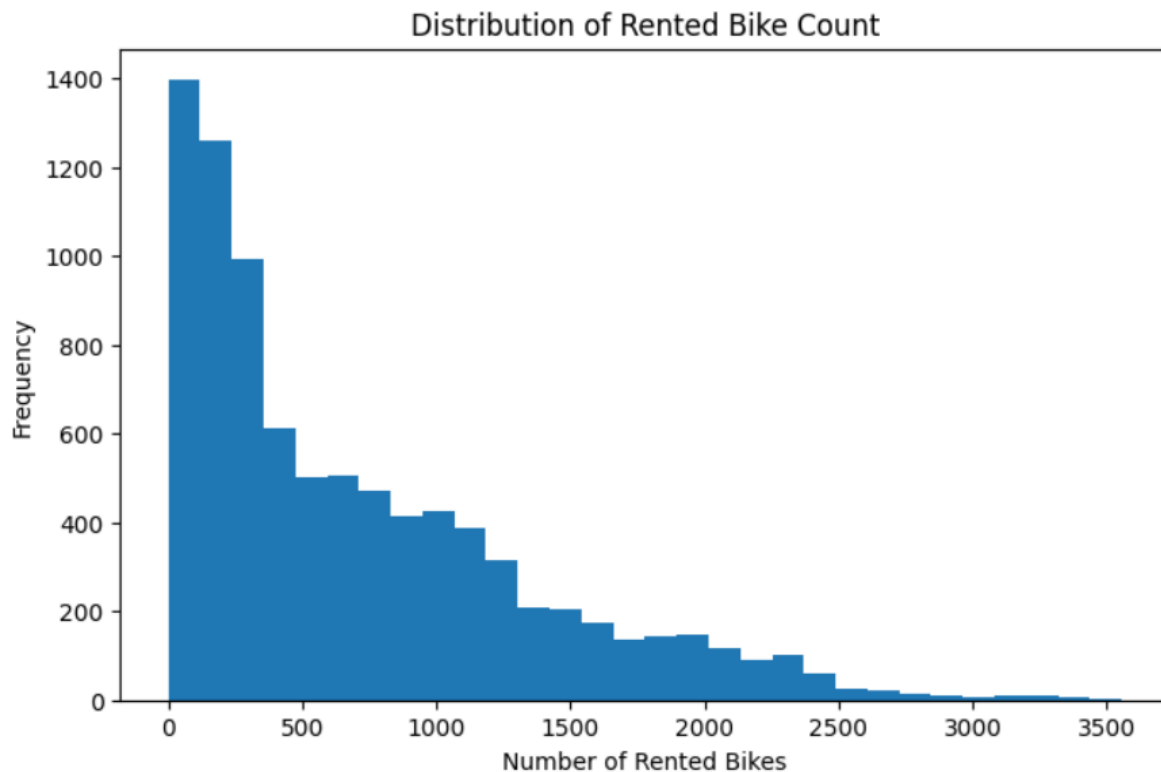
6.1 Descriptive Statistics:

Descriptive statistical measures such as mean, median, standard deviation, and quartiles were calculated for the numerical variables.

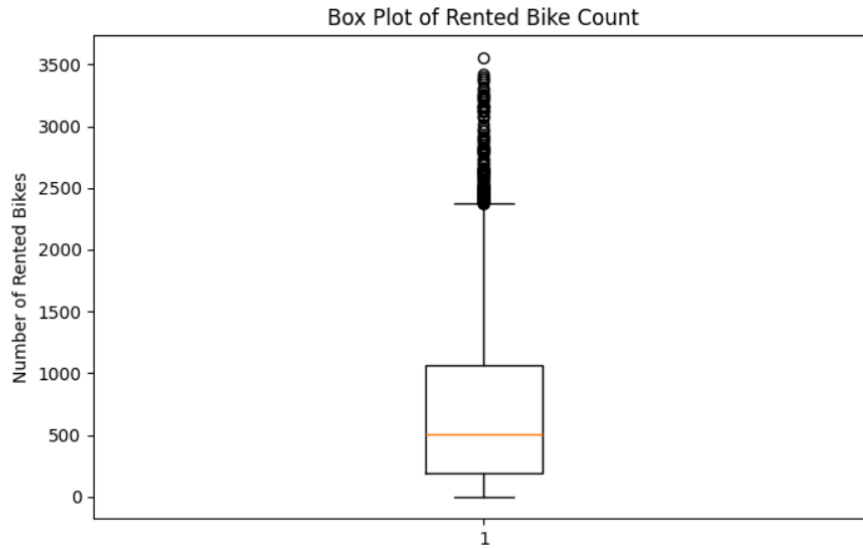
For Rented Bike Count, the mean was 764.60, the median was 504.50, and the standard deviation was 645.00. The difference between the mean and median indicates that the distribution of bike rental demand is not perfectly symmetric.

6.2 Distribution of Bike Rental Demand:

A histogram and box plot were used to study the distribution of Rented Bike Count. These visualizations helped identify the spread of bike rental demand and the presence of unusually high or low observations.



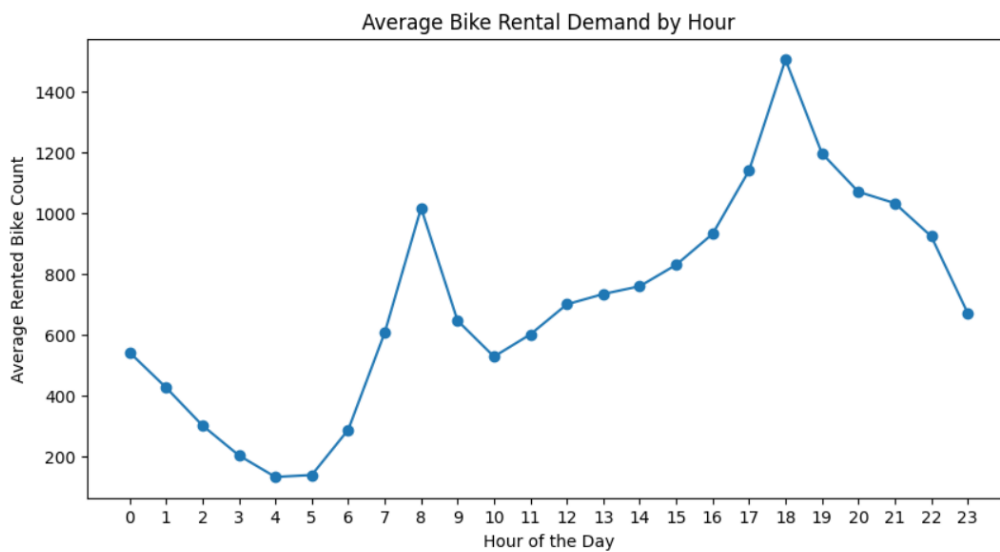
Histogram for Distribution of Rented Bike Count



Box plot of Rented Bike Count

6.3 Bike Rental Demand by Hour:

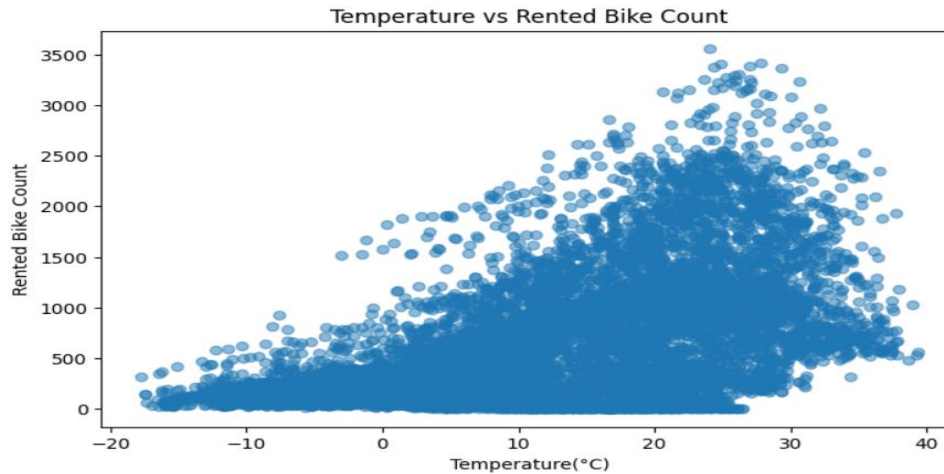
The relationship between the hour of the day and bike rental demand was examined to understand how rental demand changes throughout the day.



Line plot of Average Bike Rental Demand by Hour

6.4 Temperature and Bike Rental Demand:

A scatter plot was created to study the relationship between temperature and Rented Bike Count. Temperature showed a positive correlation with bike rental demand, indicating that higher temperatures were generally associated with higher rental demand.

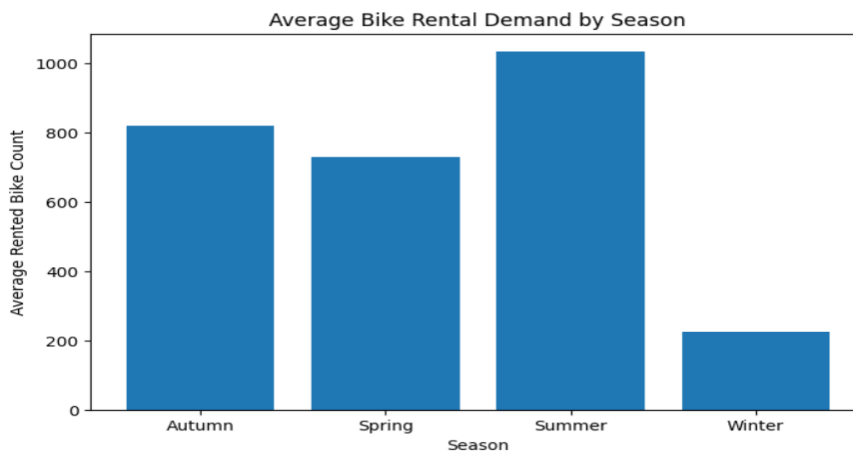


Scatter plot of Temperature vs Rented Bike Count

6.5 Bike Rental Demand by Season

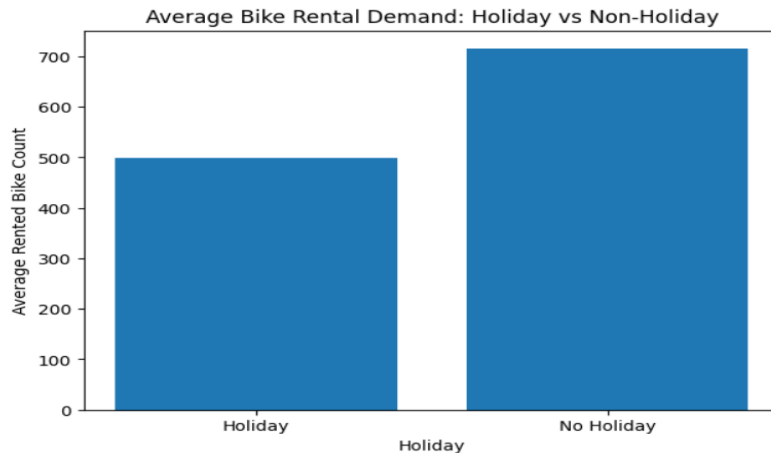
Bike rental demand was analyzed across different seasons to understand whether seasonal changes affect the number of rented bikes. A comparison of the average rental demand for each season was performed using a bar graph.

The analysis shows that bike rental demand varies across different seasons. This indicates that seasonal conditions can influence the demand for rental bikes. Factors such as temperature, weather conditions, and seasonal changes may contribute to these differences.



6.6 Holiday vs Non-Holiday Bike Rental Demand

The bike rental demand was compared between holidays and non-holidays to examine whether holiday status has an effect on rental demand. The comparison was represented using a graph showing the average number of rented bikes for both categories.

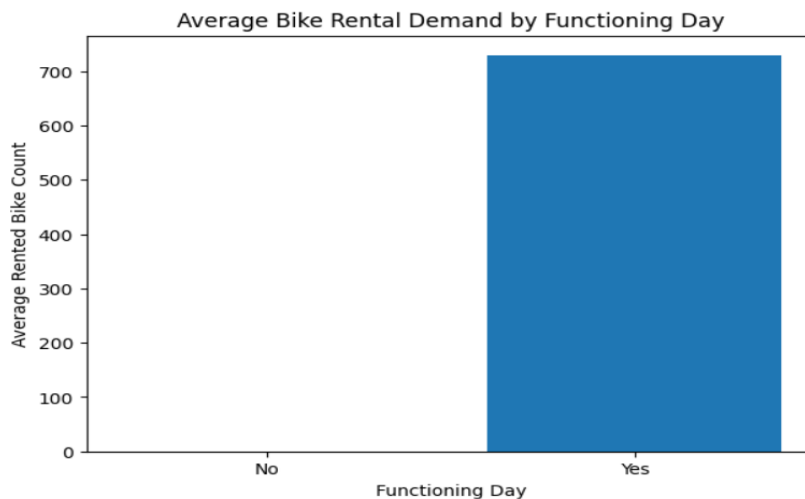


Average Bike Rental Demand: Holiday vs Non-Holiday

6.7 Functioning Day vs Non-Functioning Day

Bike rental demand was also analyzed based on whether the day was a Functioning Day or a Non-Functioning Day. A functioning day represents a day on which the bike rental service was operating normally.

The comparison helps to understand how the operation status of the service is related to the number of rented bikes. The graph shows the difference in rental demand between functioning and non-functioning days.



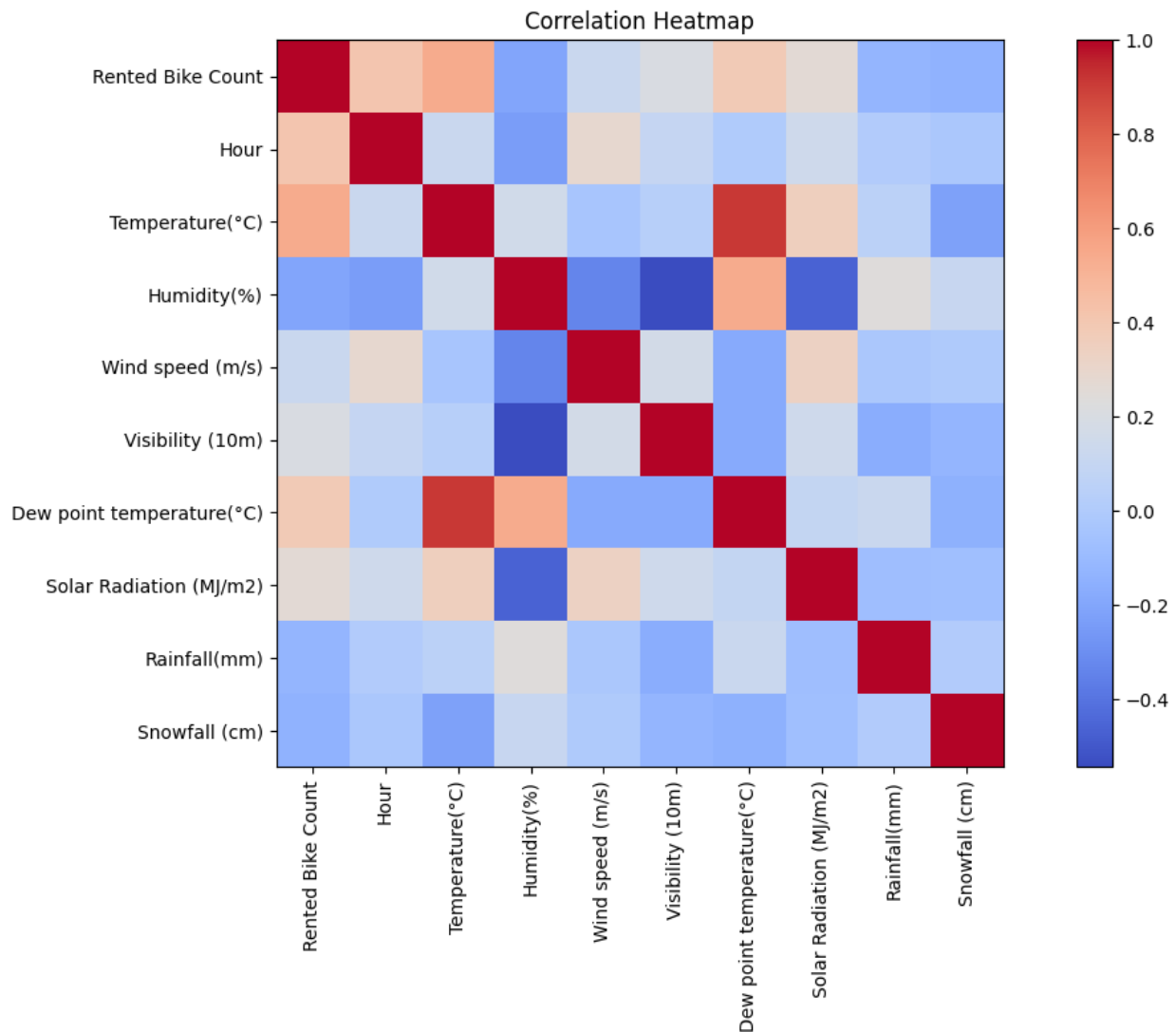
Average Bike Rental Demand by Functioning Day

6.8 Correlation Analysis:

Correlation analysis was performed to examine the relationships between numerical variables. Temperature had the strongest positive correlation with Rented Bike Count among the numerical variables, with a correlation coefficient of approximately 0.539. Hour also showed a positive correlation of approximately 0.410.

Humidity showed a negative correlation of approximately -0.200 with bike rental demand, while rainfall and snowfall also showed negative correlations.

A correlation heatmap was created to visually represent the relationships between the numerical variables.



Correlation heatmap

7. Statistical Analysis

Statistical analysis was performed to summarize the dataset and understand the relationships between bike rental demand and the numerical predictor variables. Measures of central tendency, dispersion, quartiles, and correlation were used.

7.1 Measures of Central Tendency:

The mean and median of Rented Bike Count were calculated.

Mean: 764.60 bikes & Median: 504.50 bikes

The mean is higher than the median, indicating that some observations have relatively high bike rental demand.

7.2 Measure of Dispersion:

The standard deviation of Rented Bike Count was calculated to understand the variation in hourly rental demand. Standard Deviation: 645.00 bikes

The relatively large variation indicates that the number of bikes rented differs considerably across different hours.

7.3 Quartile Analysis:

Quartiles were calculated to understand how the bike rental observations are distributed across different parts of the dataset. The first quartile (Q1), median (Q2), and third quartile (Q3) were used to describe the distribution of Rented Bike Count.

7.4 Correlation Analysis:

Pearson correlation was used to examine the relationship between Rented Bike Count and the numerical variables.

- The main correlations with bike rental demand were:

Variable	Correlation with Rented Bike Count
Temperature	0.539
Hour	0.410
Dew point temperature	0.398
Solar Radiation	0.262
Visibility	0.199
Wind speed	0.121
Rainfall	-0.123
Snowfall	-0.142
Humidity	-0.200

Temperature had the strongest positive correlation with bike rental demand among the numerical variables. Humidity, rainfall, and snowfall showed negative correlations with rental demand.

These statistical findings were considered while developing the Linear Regression model for predicting hourly bike rental demand.

8. Model Building

The objective of model building is to develop a supervised machine learning model that can predict the number of bikes rented during a particular hour.

Since the target variable, Rented Bike Count, is numerical and the problem involves predicting a continuous value, Linear Regression was selected as the supervised learning algorithm.

8.1 Selection of Variables:

The following nine numerical variables were selected as predictor variables:

- Hour
- Temperature
- Humidity
- Wind speed
- Visibility
- Dew point temperature
- Solar Radiation
- Rainfall
- Snowfall

The target variable was Rented Bike Count.

8.2 Splitting the Dataset:

The dataset was divided into training and testing datasets using an 80:20 split.

- 80% of the observations were used for training the model.
- 20% of the observations were used for testing the model.

The training data was used to learn the relationship between the predictor variables and bike rental demand, while the testing data was used to evaluate the model's performance on previously unseen observations.

8.3 Linear Regression Model:

A Linear Regression model was created using the selected predictor variables. The model was trained using the training dataset.

The trained model was then used to predict the number of rented bikes for the observations in the testing dataset.

The predicted values were compared with the actual Rented Bike Count values during model evaluation.

9. Model Evaluation

The performance of the Linear Regression model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. These measures were calculated using the actual and predicted bike rental values from the testing dataset.

9.1 Evaluation Results

Evaluation Measure	Result
Mean Absolute Error (MAE)	349.28
Mean Squared Error (MSE)	222,880.72
Root Mean Squared Error (RMSE)	472.10
R^2 Score	0.4651

9.2 Interpretation of Evaluation Measures

Mean Absolute Error (MAE) measures the average absolute difference between the actual and predicted values. The obtained MAE of 349.28 indicates that the model's predictions differed from the actual bike rental count by approximately 349 bikes on average.

Mean Squared Error (MSE) calculates the average squared difference between actual and predicted values. The MSE obtained by the model was 222,880.72.

Root Mean Squared Error (RMSE) is the square root of MSE and gives the error in the same unit as the target variable. The obtained RMSE was 472.10 bikes.

The R^2 Score indicates the proportion of variation in bike rental demand explained by the model. The model obtained an R^2 score of 0.4651, meaning that approximately 46.51% of the variation in bike rental demand was explained by the selected predictor variables.

An Actual vs Predicted graph was also created to visually compare the actual bike rental values with the values predicted by the model.

10. Interpretation of Results

The analysis was performed to understand the factors associated with hourly bike rental demand and to develop a model for predicting the number of rented bikes.

The descriptive statistical analysis showed that the average hourly bike rental demand was 764.60 bikes, while the median was 504.50 bikes. The standard deviation was 645.00, indicating considerable variation in the number of bikes rented across different hours.

The correlation analysis showed that Temperature had the strongest positive correlation with Rented Bike Count among the numerical variables, with a correlation coefficient of 0.539. Hour also showed a positive correlation of 0.410. Dew point temperature and Solar Radiation showed positive correlations of 0.398 and 0.262 respectively.

Humidity showed a negative correlation of -0.200 with bike rental demand. Rainfall and Snowfall also showed negative correlations of -0.123 and -0.142 respectively. These results indicate that weather conditions and time-related factors are associated with variations in bike rental demand.

The Linear Regression model was trained using nine numerical predictor variables. The model achieved an MAE of 349.28 and an RMSE of 472.10. The R^2 score was 0.4651, indicating that the model explained approximately 46.51% of the variation in bike rental demand.

The Actual vs Predicted graph showed a positive relationship between the actual and predicted values. However, the points were spread around the reference line, indicating that the model did not predict every observation accurately.

Overall, the analysis indicates that factors such as temperature, hour, humidity, and other weather conditions are associated with bike rental demand. The Linear Regression model was able to capture part of the variation in demand, but other factors may also.

11. Conclusion and Limitations

11.1 Conclusion:

This project focused on predicting hourly bike rental demand using the Seoul Bike Sharing Demand dataset. Statistical analysis and exploratory data analysis were performed to understand the distribution of bike rental demand and the relationships between demand and different weather and time-related variables. The analysis showed that temperature and hour had relatively stronger positive relationships with bike rental demand, while humidity, rainfall, and snowfall showed negative relationships.

A Linear Regression model was developed using nine numerical predictor variables. The model achieved an R^2 score of 0.4651, indicating that approximately 46.51% of the variation in bike rental demand was explained by the selected variables. The model was also evaluated using MAE, MSE, and RMSE.

The project demonstrates how statistical analysis, data visualization, and supervised machine learning can be combined to study and predict bike rental demand. The results also show that bike rental demand is affected by multiple factors and that further analysis could improve prediction performance.

11.2 Limitations:

The project has the following limitations:

1. The Linear Regression model used only selected numerical variables. Categorical variables such as Seasons, Holiday, and Functioning Day were not included in the initial model.
2. Bike rental demand may be influenced by additional factors that are not included in the dataset.
3. Linear Regression assumes a linear relationship between the predictors and the target variable, whereas bike rental demand may have more complex relationships.
4. The model produced prediction errors, as shown by the MAE, RMSE, and the spread of points in the Actual vs Predicted graph.
5. The model was developed using historical data, so its predictions may not represent future demand under conditions that differ from the dataset.